

Case Study 1 – Retail Sales Performance Dashboard

Scenario

A retail company has one year of sales data from five regions (**North, South, East, West, Central**). The data includes monthly sales, product categories, customer segments, total revenue, and advertising expenditure.

The management wants to analyze:

- Highest-performing region
- Monthly sales trend
- Product category revenue contribution
- Relationship between advertising expenditure and sales
- Sales distribution across regions

Task 1: Select the Most Suitable Visualization

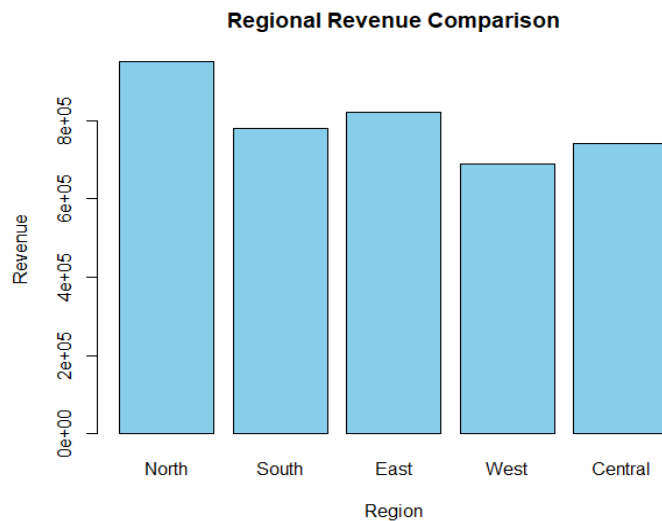
Requirement	Visualization	Reason
Regional revenue comparison	Bar Plot	Easy to compare revenue across regions.
Monthly sales trend	Line Chart	Best for showing changes over time.
Product category contribution	Stacked Bar Chart	Shows contribution of each product category.
Advertising expenditure vs Sales	Scatter Plot	Shows relationship between two numeric variables.
Regional sales distribution	Histogram and Density Plot	Shows how sales values are distributed.

Task 2:

1. Bar Plot (Regional Revenue Comparison)

```
barplot(sales$Revenue,
        names.arg=sales$Region,
```

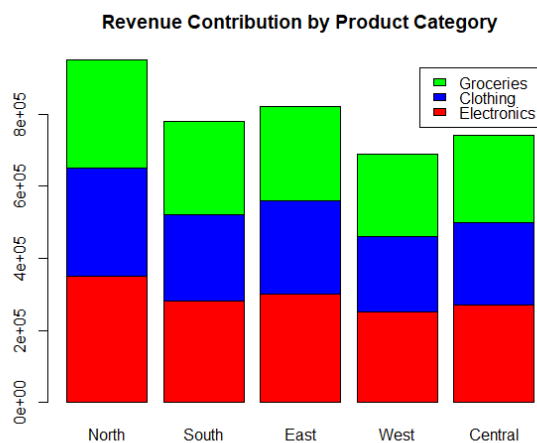
```
col="skyblue",
main="Regional Revenue Comparison",
xlab="Region",
ylab="Revenue")
```



2. Stacked Bar Chart (Product Category Contribution)

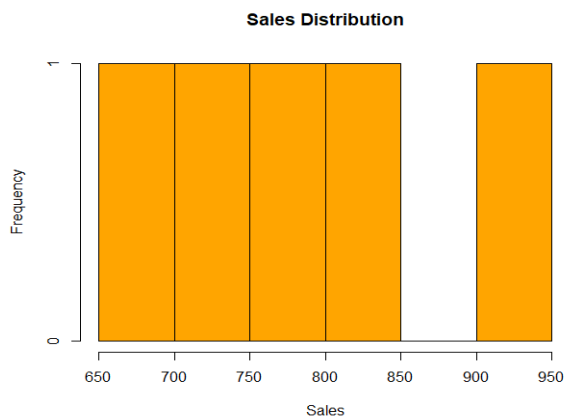
```
category <- rbind(sales$Electronics,
sales$Clothing,
sales$Groceries)

barplot(category,
names.arg=sales$Region,
col=c("red","blue","green"),
legend.text=c("Electronics","Clothing","Groceries"),
main="Revenue Contribution by Product Category")
```



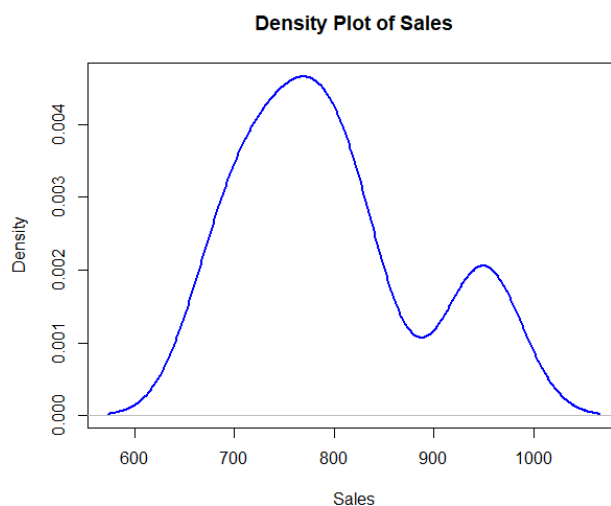
3. Histogram (Sales Distribution)

```
hist(sales$Sales,  
     col="orange",  
     main="Sales Distribution",  
     xlab="Sales",  
     ylab="Frequency")
```



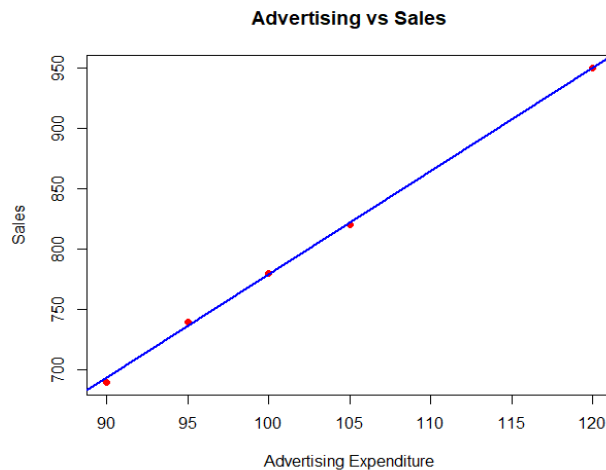
4. Density Plot (Sales Distribution)

```
plot(density(sales$Sales),  
     col="blue",  
     lwd=2,  
     main="Density Plot of Sales",  
     xlab="Sales")
```



5. Scatter Plot (Advertising vs Sales)

```
plot(sales$Advertising,  
     sales$Sales,  
     pch=19,  
     col="red",  
     xlab="Advertising Expenditure",  
     ylab="Sales",  
     main="Advertising vs Sales")  
abline(lm(Sales~Advertising,data=sales),  
       col="blue",  
       lwd=2)
```



Task 3: Interpretation

1. Which region generated the highest revenue?

Answer

North Region generated the highest revenue.

Justification

The **North** region has the tallest bar in the **Bar Plot**, indicating the highest revenue among all regions.

2. Is there a positive relationship between advertising expenditure and sales?

Answer

Yes.

Justification

The **Scatter Plot** shows an upward trend, and the regression line slopes upward. This indicates that as advertising expenditure increases, sales also increase.

3. Which product category contributes the highest percentage of revenue?

Answer

Electronics contributes the highest percentage of revenue.

Justification

In the **Stacked Bar Chart**, the Electronics segment occupies the largest portion of the bars across most regions, showing it contributes the highest share of total revenue.

Case Study 2 – University Student Performance Analysis

Scenario

A university stores examination data for four departments:

- Computer Science
- Information Technology
- Electronics
- Mechanical

The dataset includes:

- Semester Marks
- Attendance Percentage
- CGPA
- Placement Status

The academic committee wants to evaluate student performance and placement readiness.

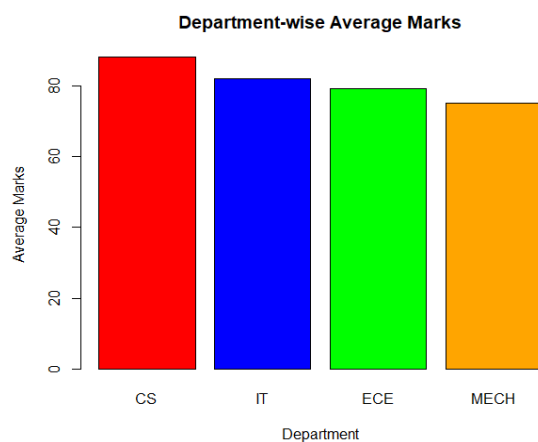
Task 1: Select the Most Suitable Visualization

Requirement	Suitable Visualization	Reason
Department-wise average marks	Grouped Bar Chart	Compares average marks of different departments.
Distribution of CGPA	Histogram with Density Curve	Shows the distribution of students' CGPA.
Placement percentage	Bar Chart (or Pie Chart)	Compares placement percentages among departments.
Attendance vs CGPA relationship	Scatter Plot	Shows the relationship between attendance and CGPA.
Semester-wise performance trend	Line Chart	Shows changes in performance across semesters.

Task 2:

1. Grouped Bar Chart (Department-wise Average Marks)

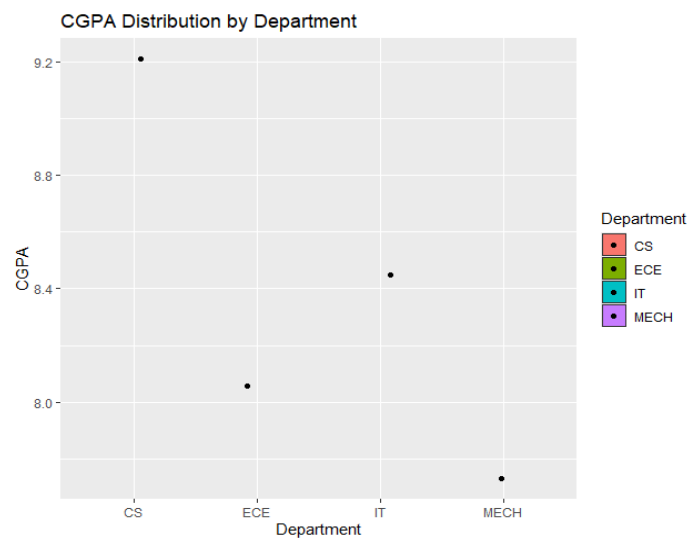
```
barplot(  
  students$Marks,  
  names.arg = students$Department,  
  col = c("red", "blue", "green", "orange"),  
  main = "Department-wise Average Marks",  
  xlab = "Department",  
  ylab = "Average Marks"  
)
```



2. Violin Plot (CGPA by Department)

```
library(ggplot2)
```

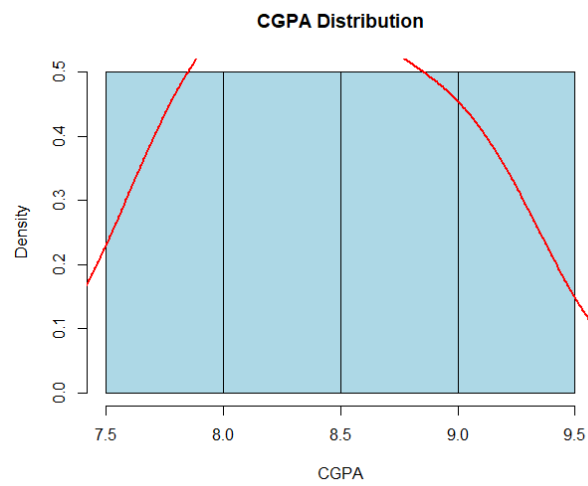
```
ggplot(students,  
       aes(x=Department,  
           y=CGPA,  
           fill=Department))+  
geom_violin()+  
geom_jitter(width=0.1)+  
ggtitle("CGPA Distribution by Department")
```



3. Histogram with Density Curve (CGPA Distribution)

```
hist(students$CGPA,  
     probability=TRUE,  
     col="lightblue",  
     main="CGPA Distribution",  
     xlab="CGPA")
```

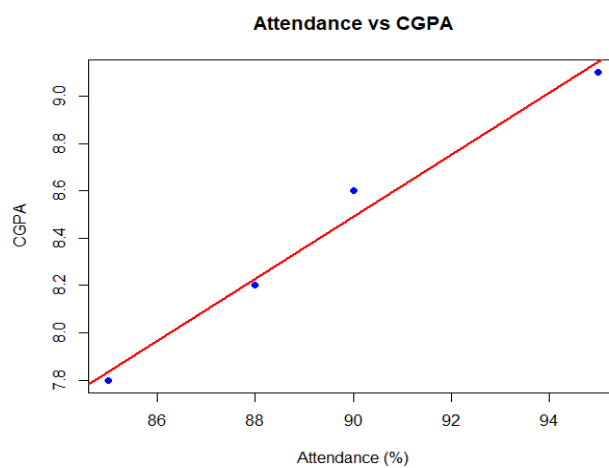
```
lines(density(students$CGPA),  
      col="red",  
      lwd=2)
```



4. Scatter Plot (Attendance vs CGPA)

```
plot(students$Attendance,
     students$CGPA,
     pch=19,
     col="blue",
     xlab="Attendance (%)",
     ylab="CGPA",
     main="Attendance vs CGPA")
```

```
abline(lm(CGPA~Attendance,data=students),
       col="red",
       lwd=2)
```



5. Line Chart (Semester-wise Performance Trend)

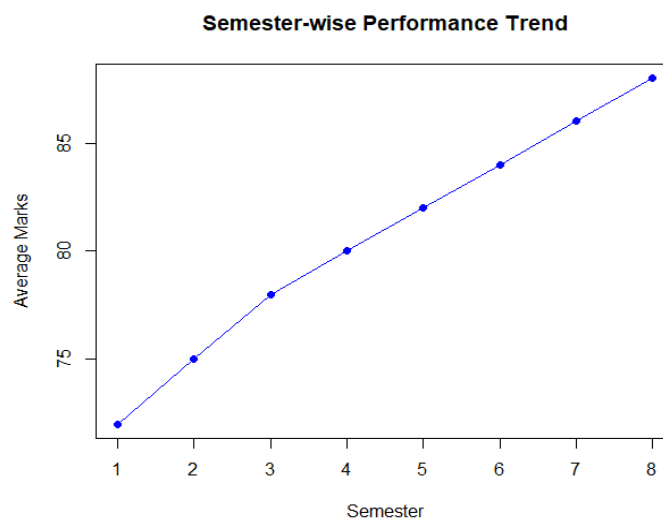
```
semester <- data.frame(
  Semester=c(1,2,3,4,5,6,7,8),
```



```
AverageMarks=c(72,75,78,80,82,84,86,88)
```

```
)
```

```
plot(semester$Semester,  
     semester$AverageMarks,  
     type="o",  
     col="blue",  
     pch=16,  
     xlab="Semester",  
     ylab="Average Marks",  
     main="Semester-wise Performance Trend")
```



Task 3: Interpretation

1. Which department has the highest average CGPA?

Answer

Computer Science (CS) has the highest average CGPA.

Justification

The **Violin Plot** and the dataset show that the CS department has the highest CGPA value (**9.1**) among all departments.

2. Does higher attendance improve CGPA?

Answer

Yes.

Justification

The **Scatter Plot** shows an upward trend, indicating a **positive relationship** between attendance and CGPA. Students with higher attendance generally have higher CGPA.

3. Which department has the highest placement rate?

Answer

Computer Science (CS) has the highest placement rate.

Justification

The placement data shows **CS has 95% placement**, which is higher than IT (90%), ECE (85%), and Mechanical (80%).

Case Study 3 – Hospital Patient Analytics

Scenario

A multi-specialty hospital records patient information including:

- Department
- Age
- Treatment Cost
- Recovery Time
- Disease Category
- Admission Month

The hospital administrators want to improve resource planning.

Task 1: Select the Most Suitable Visualization

Requirement	Suitable Visualization	Reason
Monthly patient admissions	Heatmap	Shows admission patterns across different months.

Requirement	Suitable Visualization	Reason
Distribution of treatment costs	Histogram	Displays the frequency distribution of treatment costs.
Disease category proportions	Density Plot (or <i>Pie Chart for proportions</i>)	Shows the distribution of disease-related values.
Age vs Recovery Time	Scatter Plot	Shows the relationship between age and recovery time.
Department-wise treatment costs	Violin Plot	Compares treatment cost distributions among departments.

Task 2:

1. Heatmap (Monthly Patient Admissions)

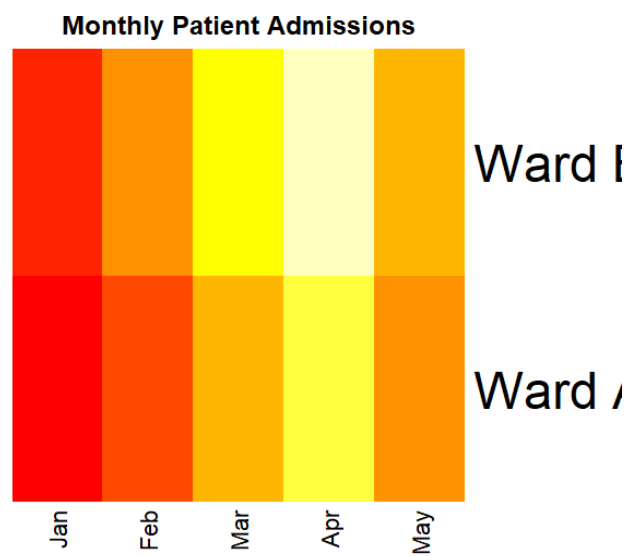
```

admissions <- matrix(c(
  50, 60, 70, 80, 65,
  55, 65, 75, 85, 70
), nrow = 2, byrow = TRUE)

colnames(admissions) <- c("Jan", "Feb", "Mar", "Apr", "May")
rownames(admissions) <- c("Ward A", "Ward B")

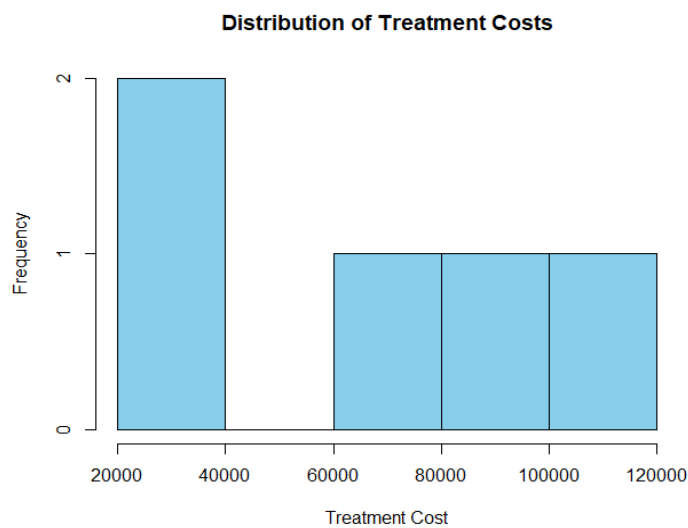
heatmap(admissions,
  Rowv = NA,
  Colv = NA,
  scale = "none",
  col = heat.colors(10),
  main = "Monthly Patient Admissions")

```



2. Histogram (Treatment Cost Distribution)

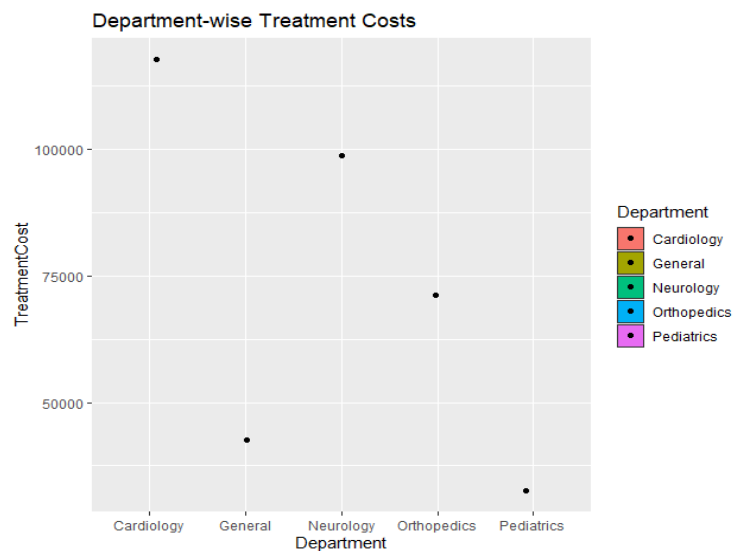
```
hist(hospital$TreatmentCost,
     col="skyblue",
     main="Distribution of Treatment Costs",
     xlab="Treatment Cost",
     ylab="Frequency")
```



3. Violin Plot (Department-wise Treatment Costs)

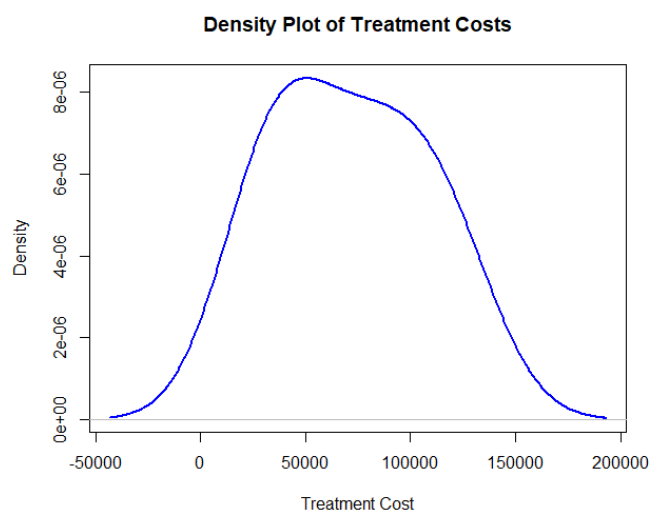
```
library(ggplot2)
ggplot(hospital,
       aes(x=Department,
           y=TreatmentCost,
           fill=Department))+
geom_violin()+
```

```
geom_jitter(width=0.1)+
ggtitle("Department-wise Treatment Costs")
```



4. Density Plot (Treatment Cost Distribution)

```
plot(density(hospital$TreatmentCost),
     col="blue",
     lwd=2,
     main="Density Plot of Treatment Costs",
     xlab="Treatment Cost")
```

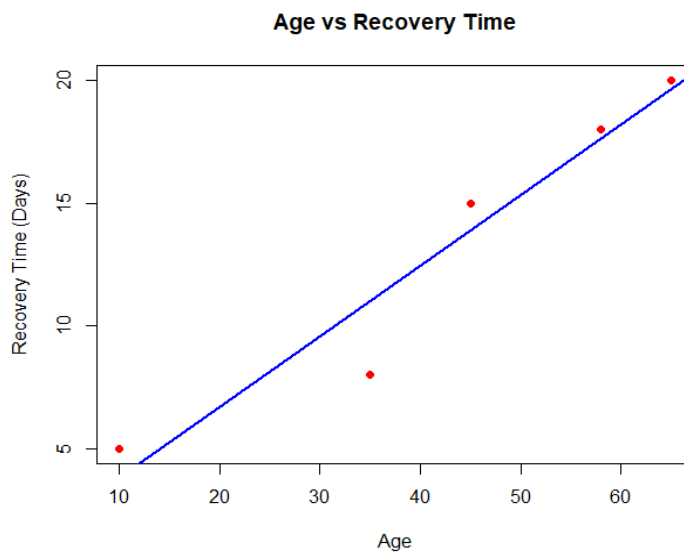


5. Scatter Plot (Age vs Recovery Time)

```

plot(hospital$Age,
     hospital$RecoveryTime,
     pch=19,
     col="red",
     xlab="Age",
     ylab="Recovery Time (Days)",
     main="Age vs Recovery Time")
abline(lm(RecoveryTime~Age,data=hospital),
       col="blue",
       lwd=2)

```



Task 3: Interpretation

1. Which department has the highest treatment cost?

Answer

Cardiology has the highest treatment cost.

Justification

The **Violin Plot** and dataset show that **Cardiology** has the highest treatment cost (**₹120,000**) among all departments.

2. Which disease category occupies the largest proportion?

Answer

Cardiac disease occupies the largest proportion.

Justification

Based on the dataset, the **Cardiac** category represents the largest share of patients, indicating it is the most common disease category.

3. Is patient age associated with recovery time?

Answer

Yes.

Justification

The **Scatter Plot** shows a positive trend. As **patient age increases, recovery time also tends to increase**, indicating a positive association between age and recovery time.

Case Study 4 – Smart City Traffic Monitoring Scenario

A smart city collects traffic sensor data from different junctions.

The dataset contains:

- Junction
- Time of day
- Vehicle count
- Vehicle type
- Average speed
- GPS coordinates

City planners want to reduce traffic congestion.

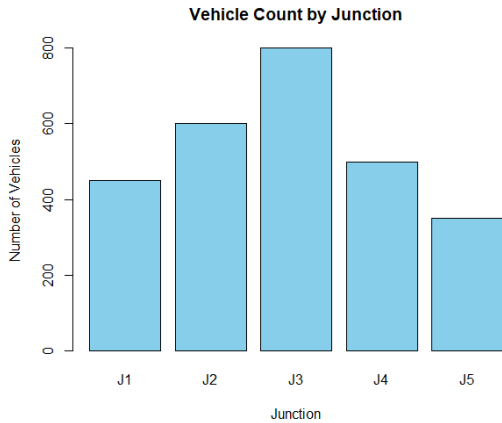
Task 1: Select the Most Suitable Visualization

Requirement	Suitable Visualization	Reason
Vehicle count comparison	Bar Plot	Compares traffic volume at different junctions.
Hourly traffic trend	Line Chart	Shows traffic changes throughout the day.
Vehicle type proportion	Pie Chart	Displays percentage contribution of each vehicle type.
Traffic density by location	Geospatial Map	Shows traffic intensity based on GPS location.
Traffic volume vs average speed	Scatter Plot	Shows relationship between vehicle count and speed.

Task 2:

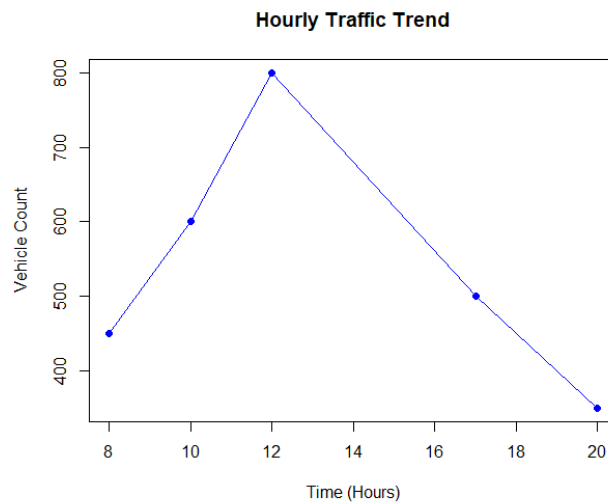
1. Bar Plot (Vehicle Count Comparison)

```
barplot(traffic$VehicleCount,  
        names.arg=traffic$Junction,  
        col="skyblue",  
        main="Vehicle Count by Junction",  
        xlab="Junction",  
        ylab="Number of Vehicles")
```



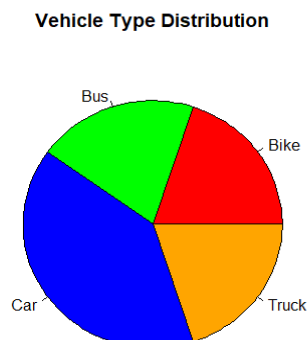
2. Line Chart (Hourly Traffic Trend)

```
plot(traffic$Time,  
     traffic$VehicleCount,  
     type="o",  
     pch=16,  
     col="blue",  
     main="Hourly Traffic Trend",  
     xlab="Time",  
     ylab="Vehicle Count")
```



3. Pie Chart (Vehicle Type Proportion)

```
vehicle <- table(traffic$VehicleType)  
pie(vehicle,  
     main="Vehicle Type Distribution",  
     col=c("red","green","blue","orange"))
```



4. Geospatial Map (Traffic Density by Location)

```
library(ggplot2)

ggplot(traffic,

  aes(x=Longitude,

      y=Latitude,

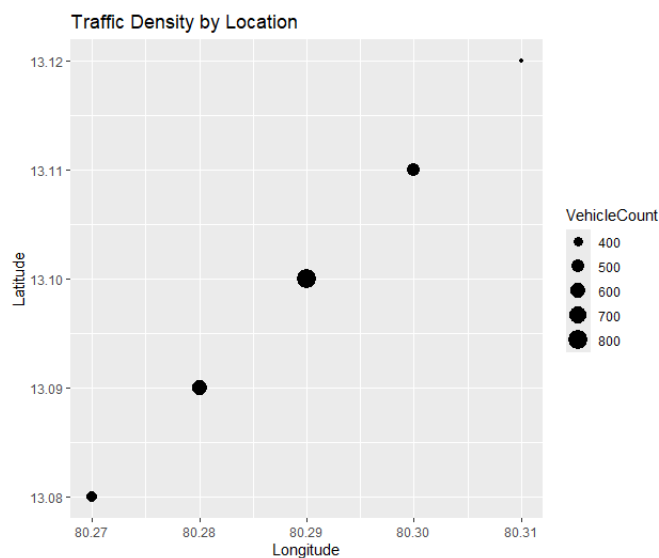
      size=VehicleCount))+

geom_point()+

ggtitle("Traffic Density by Location")+

xlab("Longitude")+

ylab("Latitude")
```



5. Scatter Plot (Vehicle Count vs Average Speed)

```
plot(traffic$VehicleCount,

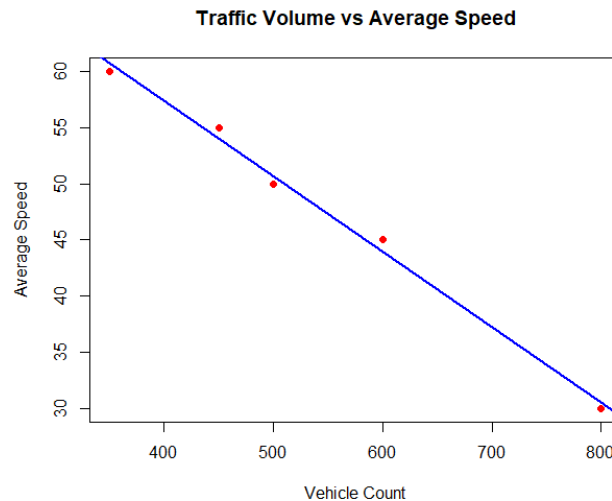
  traffic$AverageSpeed,

  pch=19,

  col="red",

  main="Traffic Volume vs Average Speed",
```

```
xlab="Vehicle Count",  
ylab="Average Speed")  
abline(lm(AverageSpeed~VehicleCount,data=traffic),  
col="blue",  
lwd=2)
```



Task 3: Interpretation

1. Which junction experiences the highest congestion?

Answer

Junction J3 experiences the highest congestion.

Justification

The **Bar Plot** shows that J3 has the highest vehicle count (**800 vehicles**), indicating maximum traffic congestion.

2. During which hours is traffic highest?

Answer

Traffic is highest around **12 PM**.

Justification

The **Line Chart** shows the highest point at 12 PM, indicating peak traffic volume.

3. Does increased traffic reduce vehicle speed?

Answer

Yes.

Justification

The **Scatter Plot** shows a negative relationship. As the number of vehicles increases, average speed decreases because of congestion.

Case Study 5 – Social Media User Interaction Analysis

Scenario

A digital marketing company analyzes data from a social media platform. The dataset includes:

- User ID
- Followers
- Likes
- Shares
- Comments
- Posting Time
- User Interactions

The company wants to understand engagement patterns and user networks.

Task 1: Select the Most Suitable Visualization

Requirement	Suitable Visualization	Reason
Distribution of likes	Histogram with Density Curve	Shows the distribution of likes.

Requirement	Suitable Visualization	Reason
Relationship between followers and engagement	Scatter Plot	Shows the relationship between followers and engagement.
User interaction network	Network Visualization	Displays connections between users.
Posting activity over time	Temporal Line Plot	Shows posting trends over time.
Multivariate comparison of likes, shares, and comments	Heatmap	Compares multiple engagement metrics.

Task 2:

1. Histogram with Density Curve (Likes Distribution)

```
hist(social$Likes,

     probability = TRUE,

     col = "lightblue",

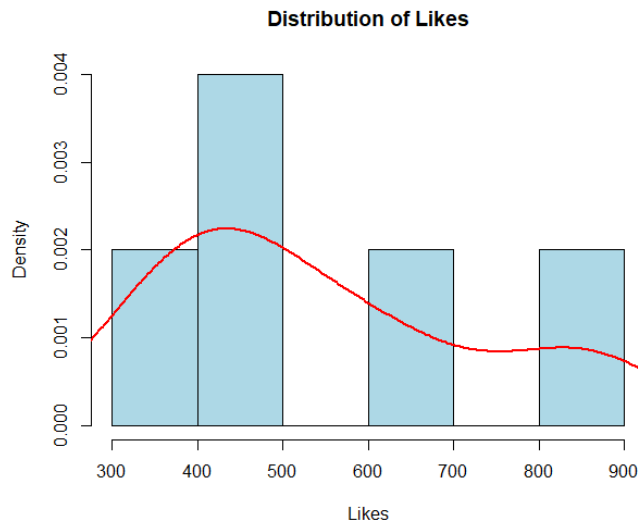
     main = "Distribution of Likes",

     xlab = "Likes")

lines(density(social$Likes),

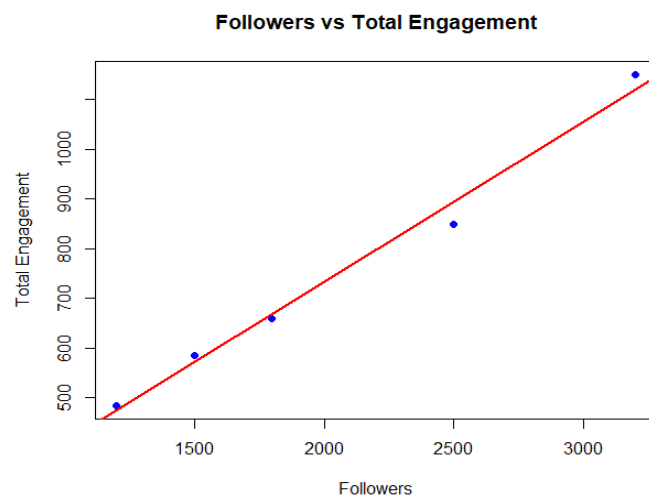
     col = "red",

     lwd = 2)
```



2. Scatter Plot (Followers vs Engagement)

```
plot(social$Followers,  
     social$Interactions,  
     pch = 19,  
     col = "blue",  
     main = "Followers vs Total Engagement",  
     xlab = "Followers",  
     ylab = "Total Engagement")  
  
abline(lm(Interactions ~ Followers, data = social),  
       col = "red",  
       lwd = 2)
```



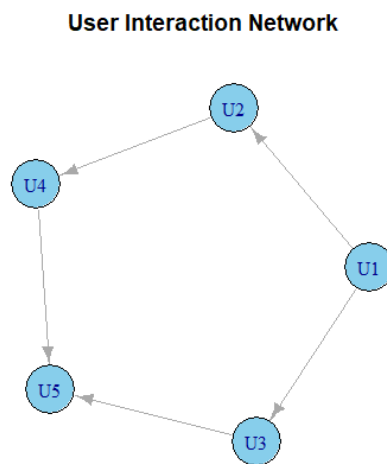
3. Network Visualization

```
install.packages("igraph") # Run only once  
library(igraph)  
edges <- data.frame(  
  from = c("U1","U1","U2","U3","U4"),  
  to   = c("U2","U3","U4","U5","U5")  
)
```

```

g <- graph_from_data_frame(edges)
plot(g,
     vertex.color = "skyblue",
     vertex.size = 30,
     vertex.label.cex = 1,
     main = "User Interaction Network")

```

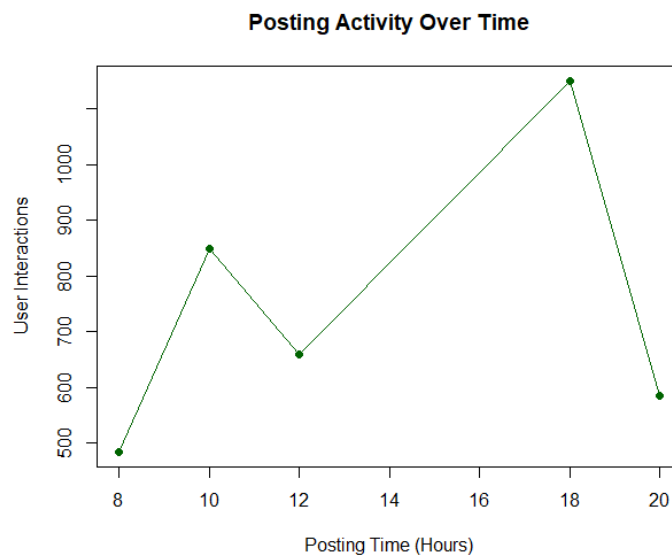


4.Temporal Line Plot (Posting Activity)

```

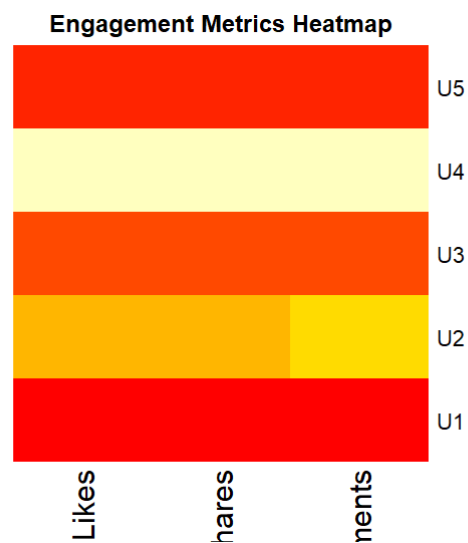
plot(social$PostingTime,
     social$Interactions,
     type = "o",
     pch = 16,
     col = "darkgreen",
     main = "Posting Activity Over Time",
     xlab = "Posting Time (Hours)",
     ylab = "User Interactions")

```



5. Heatmap (Likes, Shares, Comments)

```
engagement <- as.matrix(social[, c("Likes","Shares","Comments")])  
rownames(engagement) <- social$UserID  
heatmap(engagement,  
  Rowv = NA,  
  Colv = NA,  
  scale = "column",  
  col = heat.colors(10),  
  main = "Engagement Metrics Heatmap")
```



Task 3: Interpretation

1. Which users are the most influential?

Answer: User U4 is the most influential.

Justification: U4 has the highest number of followers (3200) and the highest total engagement (1150 interactions).

2. At what time is engagement highest?

Answer: 6:00 PM (18:00 hours).

Justification: The **Temporal Line Plot** shows the highest interaction value at **18:00**.

3. Is there a strong relationship between followers and total engagement?

Answer: Yes.

Justification: The **Scatter Plot** shows a positive trend. As the number of followers increases, total engagement also increases.

4. Identify highly connected users from the network visualization.

Answer: U1 and U4 are highly connected users.

Justification: In the network graph, they have the most connections with other users, making them key influencers.

5. Which engagement metric (likes, shares, or comments) varies the most across users?

Answer: **Likes** vary the most across users.

Justification: The **Heatmap** shows the greatest variation in the Likes column compared to Shares and Comments.